

14.3 SHORT-RUN COST

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To produce more output (total product) in the short run, a firm must employ more labor, which means that it must increase its costs. We describe the relationship between output and cost using three cost concepts:

- Total cost
- Marginal cost
- Average cost

Total Cost

A firm's **total cost** (TC) is the cost of all the factors of production used by the firm. Total cost divides into two parts: total fixed cost and total variable cost. **Total fixed cost** (TFC) is the cost of a firm's fixed factors of production: land, capital, and entrepreneurship. In the short run, the quantities of these inputs don't change as output changes, so total fixed cost doesn't change as output changes. **Total variable cost** (TVC) is the cost of a firm's variable factor of production—labor. To change its output in the short run, a firm must change the quantity of labor it employs, so total variable cost changes as output changes.

Total cost is the sum of total fixed cost and total variable cost. That is,

$$TC = TFC + TVC.$$

Table 14.2 shows Sam's Smoothies' total costs. Sam's fixed costs are \$10 an hour regardless of whether it operates or not— TFC is \$10 an hour. To produce smoothies, Samantha hires labor, which costs \$6 an hour. TVC , which increases as output increases, equals the number of workers per hour multiplied by \$6. For example, to produce 6 gallons an hour, Samantha hires 3 workers, so TVC is \$18 an hour. TC is the sum of TFC and TVC . So to produce 6 gallons an hour, TC is \$28. Check the calculation in each row and note that to produce some quantities—2 gallons an hour, for example—Sam hires a worker for only part of the hour.

Total cost

The cost of all the factors of production used by a firm.

Total fixed cost

The cost of the firm's fixed factors of production—the cost of land, capital, and entrepreneurship.

Total variable cost

The cost of the firm's variable factor of production—the cost of labor.

TABLE 14.2

Sam's Smoothies' Total Costs

Labor (workers per hour)	Output (gallons per hour)	Total fixed cost	Total variable cost (dollars per hour)	Total cost	Sam's fixed factors of production are land, capital, and entrepreneurship. Total fixed cost is constant regardless of the quantity produced. Sam's variable factor of production is labor. Total variable cost is the cost of labor. Total cost is the sum of total fixed cost and total variable cost. The highlighted row shows that to produce 6 gallons of smoothies, Sam's hires 3 workers. Total fixed cost is \$10 an hour. Total variable cost is the cost of the 3 workers. At \$6 an hour, 3 workers cost \$18. Sam's total cost of producing 6 gallons an hour is \$10 plus \$18, which equals \$28.
0	0	10	0	10.00	
1.00	1	10	6.00	16.00	
1.60	2	10	9.60	19.60	
2.00	3	10	12.00	22.00	
2.35	4	10	14.10	24.10	
2.65	5	10	15.90	25.90	
3.00	6	10	18.00	28.00	
3.40	7	10	20.40	30.40	
4.00	8	10	24.00	34.00	
5.00	9	10	30.00	40.00	

Figure 14.5 illustrates Sam's total cost curves. The green total fixed cost curve (TFC) is horizontal because total fixed cost does not change when output changes. It is a constant at \$10 an hour. The purple total variable cost curve (TVC) and the blue total cost curve (TC) both slope upward because variable cost increases as output increases. The arrows highlight total fixed cost as the vertical distance between the TVC and TC curves.

Let's now look at Sam's Smoothies' marginal cost.

■ Marginal Cost

In Figure 14.5, total variable cost and total cost increase at a decreasing rate at small levels of output and then begin to increase at an increasing rate as output increases. To understand these patterns in the changes in total cost, we need to use the concept of *marginal cost*.

Marginal cost

The change in total cost that results from a one-unit increase in output.

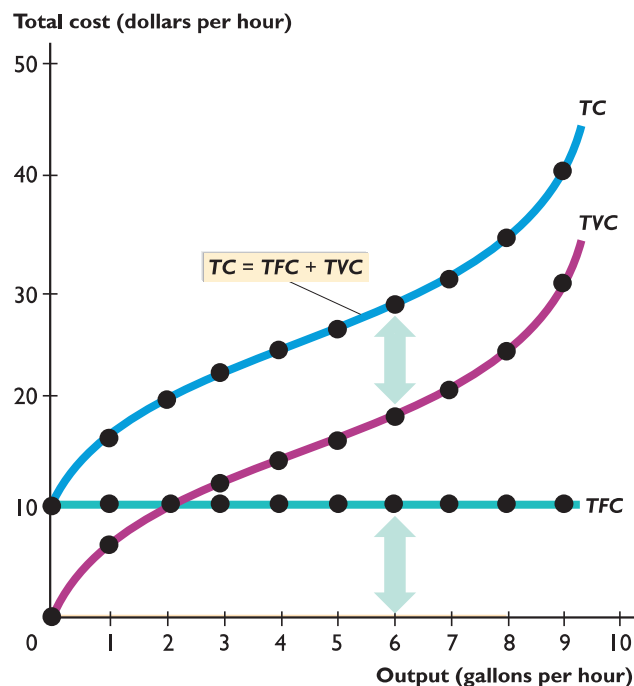
A firm's **marginal cost** is the change in total cost that results from a one-unit increase in output. Table 14.3 calculates the marginal cost for Sam's Smoothies. When, for example, output increases from 5 gallons to 6 gallons an hour, total cost increases from \$25.90 to \$28. So the marginal cost of this gallon of smoothies is \$2.10 (\$28 – \$25.90). Notice that marginal cost is located midway between the total costs to emphasize that it is the result of *changing* outputs

Marginal cost tells us how total cost changes as output changes. The final cost concept tells us what it costs, on average, to produce a unit of output. Let's now look at Sam's average costs.

FIGURE 14.5
Total Cost Curves at Sam's Smoothies

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Total fixed cost (TFC) is constant—it graphs as a horizontal line—and total variable cost (TVC) increases as output increases. Total cost (TC) also increases as output increases. The vertical distance between the total cost curve and the total variable cost curve is total fixed cost, as illustrated by the two arrows.



■ Average Cost

There are three average cost concepts:

- Average fixed cost
- Average variable cost
- Average total cost

Average fixed cost (AFC) is total fixed cost per unit of output. **Average variable cost (AVC)** is total variable cost per unit of output. **Average total cost (ATC)** is total cost per unit of output. The average cost concepts are calculated from the total cost concepts as follows:

$$TC = TFC + TVC.$$

Divide each total cost term by the quantity produced, Q , to give

$$\frac{TC}{Q} = \frac{TFC}{Q} + \frac{TVC}{Q}.$$

or

$$ATC = AFC + AVC.$$

Table 14.3 shows these average costs. For example, when output is 6 gallons an hour, AFC is $(\$10 \div 6)$, which equals \$1.67; AVC is $(\$18 \div 6)$, which equals \$3.00; and ATC is $(\$28 \div 6)$, which equals \$4.67. Note that ATC (\$4.67) equals AFC (\$1.67) plus AVC (\$3.00).

Average fixed cost

Total fixed cost per unit of output.

Average variable cost

Total variable cost per unit of output.

Average total cost

Total cost per unit of output, which equals average fixed cost plus average variable cost.

TABLE 14.3

Sam's Smoothies' Marginal Cost and Average Cost

Output (gallons per hour)	Total cost (dollars per hour)	Marginal cost (dollars per gallon)	Average fixed cost	Average variable cost	Average total cost
(dollars per gallon)					
0	10.00	6.00	—	—	—
1	16.00	3.60	10.00	6.00	16.00
2	19.60	2.40	5.00	4.80	9.80
3	22.00	2.10	3.33	4.00	7.33
4	24.10	1.80	2.50	3.53	6.03
5	25.90	2.10	2.00	3.18	5.18
6	28.00	2.40	1.67	3.00	4.67
7	30.40	3.60	1.43	2.91	4.34
8	34.00	6.00	1.25	3.00	4.25
9	40.00		1.11	3.33	4.44

To produce 6 gallons of smoothies an hour, Sam's total cost is \$28. Table 14.2 shows that this total cost is the sum of total fixed cost (\$10) and total variable cost (\$18).

Marginal cost is the increase in total cost that results from a one-unit increase in output. When Sam's increases output from 5 gallons to 6 gallons an hour, total cost increases from \$25.90 to \$28.00, an increase of \$2.10 a gallon. The marginal cost of the sixth gallon an hour is \$2.10. Marginal cost is located midway between the total costs to emphasize that it is the result of *changing* output.

When Sam's produces 6 gallons an hour, average fixed cost $(\$10 \div 6 \text{ gallons})$ is \$1.67 a gallon; average variable cost $(\$18 \div 6 \text{ gallons})$ is \$3.00 a gallon; average total cost $(\$28 \div 6 \text{ gallons})$ is \$4.67 a gallon.

Figure 14.6 graphs the marginal cost and average cost data in Table 14.3. The red marginal cost curve (MC) is U-shaped because of the way in which marginal product changes. Recall that when Samantha hires a second or a third worker, marginal product increases and output increases to 6 gallons an hour (Figure 14.3 on p. 351). Over this output range, marginal cost decreases as output increases. When Samantha hires a fourth or more workers, marginal product decreases but output increases up to 9 gallons an hour (Figure 14.3). Over this output range, marginal cost increases as output increases.

The green average fixed cost curve (AFC) slopes downward. As output increases, the same constant total fixed cost is spread over a larger output. The blue average total cost curve (ATC) and the purple average variable cost curve (AVC) are U-shaped. The vertical distance between the average total cost and average variable cost curves is equal to average fixed cost—as indicated by the two arrows. That distance shrinks as output increases because average fixed cost decreases with increasing output.

The marginal cost curve intersects the average variable cost curve and the average total cost curve at their minimum points. That is, when marginal cost is less than average cost, average cost is decreasing; and when marginal cost exceeds average cost, average cost is increasing. This relationship holds for both the ATC curve and the AVC curve and is another example of the relationship you saw in Figure 14.4 for average product and marginal product.

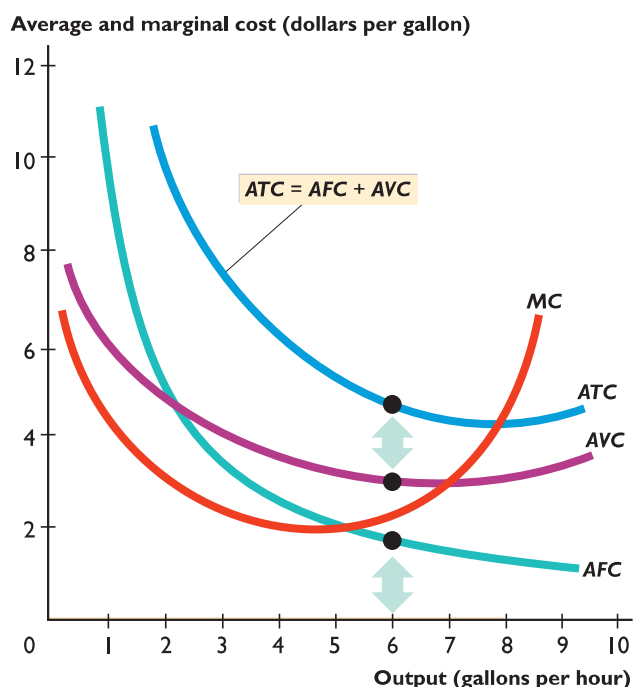
FIGURE 14.6

Average Cost Curves and Marginal Cost Curve at Sam's Smoothies

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Average fixed cost decreases as output increases. The average fixed cost curve (AFC) slopes downward. The average total cost curve (ATC) and average variable cost curve (AVC) are U-shaped. The vertical distance between these two curves is equal to average fixed cost, as illustrated by the two arrows.

Marginal cost is the change in total cost when output increases by one unit. The marginal cost curve (MC) is U-shaped and intersects the average variable cost curve and the average total cost curve at their minimum points.



■ Why the Average Total Cost Curve Is U-Shaped

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Average total cost, *ATC*, is the sum of average fixed cost, *AFC*, and average variable cost, *AVC*. So the shape of the *ATC* curve combines the shapes of the *AFC* and *AVC* curves. The U-shape of the average total cost curve arises from the influence of two opposing forces:

- Spreading total fixed cost over a larger output
- Decreasing marginal returns

When output increases, the firm spreads its total fixed costs over a larger output and its average fixed cost decreases—its average fixed cost curve slopes downward.

Decreasing marginal returns means that as output increases, ever larger amounts of labor are needed to produce an additional unit of output. So average variable cost eventually increases, and the *AVC* curve eventually slopes upward.

The shape of the average total cost curve combines these two effects. Initially, as output increases, both average fixed cost and average variable cost decrease, so average total cost decreases and the *ATC* curve slopes downward. But as output increases further and decreasing marginal returns set in, average variable cost begins to increase. Eventually, average variable cost increases more quickly than average fixed cost decreases, so average total cost increases and the *ATC* curve slopes upward.

All the short-run cost concepts that you've met are summarized in Table 14.4.

TABLE 14.4

A Compact Glossary of Costs

Term	Symbol	Definition	Equation
Fixed cost		The cost of a fixed factor of production that is independent of the quantity produced	
Variable cost		The cost of a variable factor of production that varies with the quantity produced	
Total fixed cost	<i>TFC</i>	Cost of the fixed factors of production	
Total variable cost	<i>TVC</i>	Cost of the variable factor of production	
Total cost	<i>TC</i>	Cost of all factors of production	$TC = TFC + TVC$
Marginal cost	<i>MC</i>	Change in total cost resulting from a one-unit increase in output (<i>Q</i>)	$MC = \Delta TC \div \Delta Q^*$
Average fixed cost	<i>AFC</i>	Total fixed cost per unit of output	$AFC = TFC \div Q$
Average variable cost	<i>AVC</i>	Total variable cost per unit of output	$AVC = TVC \div Q$
Average total cost	<i>ATC</i>	Total cost per unit of output	$ATC = AFC + AVC$

*In this equation, the Greek letter delta (Δ) stands for "change in."

■ Cost Curves and Product Curves

A firm's cost curves and product curves are linked, and Figure 14.7 shows how. The upper graph shows the average product curve, AP , and the marginal product curve, MP . The lower graph shows the average variable cost curve, AVC , and the marginal cost curve, MC .

As labor increases up to 2.5 workers a day (upper graph), output increases to 4 units a day (lower graph). Marginal product and average product rise and marginal cost and average variable cost fall. At the point of maximum marginal product, marginal cost is at a minimum.

As labor increases to 3.5 workers a day (upper graph), output increases to 7 units a day (lower graph). Marginal product falls and marginal cost rises, but average product continues to rise and average variable cost continues to fall. At the point of maximum average product, average variable cost is at a minimum. As labor increases further, output increases. Average product diminishes and average variable cost increases.

■ Shifts in the Cost Curves

The position of a firm's short-run cost curves, in Figures 14.5 and 14.6, depends on two factors:

- Technology
- Prices of factors of production

Technology

A technological change that increases productivity shifts the total product curve upward. It also shifts the marginal product curve and the average product curve upward. With a better technology that increases productivity, the same factors of production can produce more output, so an advance in technology lowers the average and marginal costs and shifts the short-run cost curves downward.

For example, advances in robotic technology have increased productivity in the automobile industry. As a result, the product curves of Chrysler, Ford, and GM have shifted upward, and their average and marginal cost curves have shifted downward. But the relationships between their product curves and cost curves have not changed. The curves are still linked, as in Figure 14.7.

Often a technological advance results in a firm using more capital, a fixed factor of production, and less labor, a variable factor of production. For example, today telephone companies use computers to connect long-distance calls instead of the human operators they used in the 1980s. When a telephone company makes this change, total variable cost decreases and total cost decreases, but total fixed cost increases. This change in the mix of fixed cost and variable cost means that at small output levels, average total cost might increase, but at large output levels, average total cost decreases.

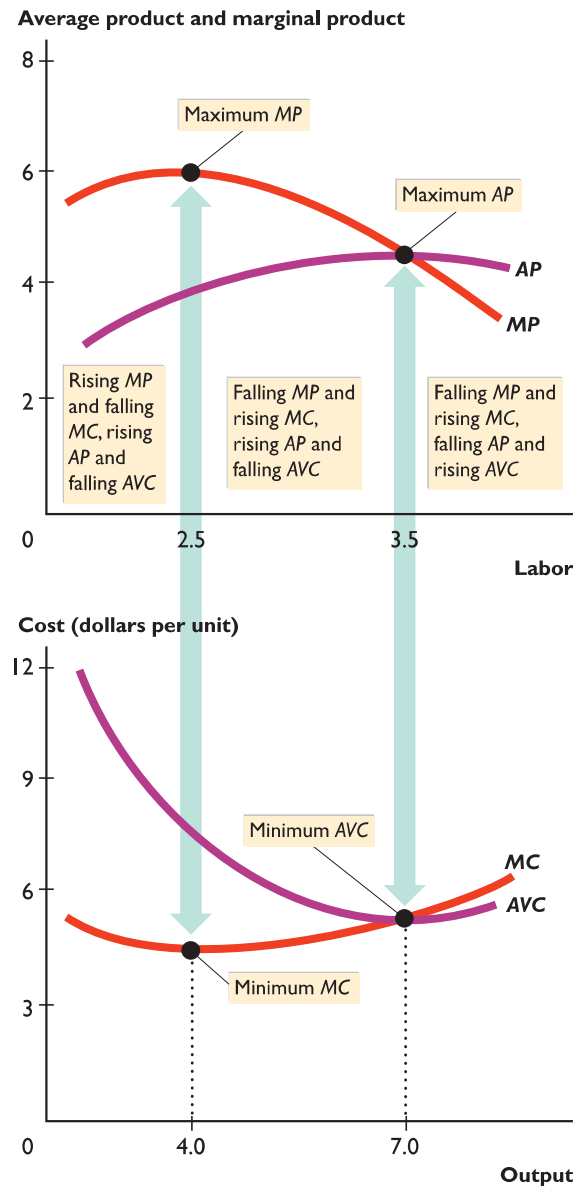
Prices of Factors of Production

An increase in the price of a factor of production increases costs and shifts the cost curves. But how the curves shift depends on which resource price changes. An increase in rent or some other component of *fixed* cost shifts the fixed cost curves (TFC and AFC) upward and shifts the total cost curve (TC) upward but leaves the variable cost curves (AVC and TVC) and the marginal cost curve (MC) unchanged.

FIGURE 14.7

Product Curves and Cost Curves

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A firm's MC curve is linked to its MP curve. If, as the firm hires more labor up to 2.5 workers a day, the firm's marginal product rises, its marginal cost falls. If marginal product is at a maximum, marginal cost is at a minimum. If, as the firm hires more labor, its marginal product diminishes, its marginal cost rises.

A firm's AVC curve is linked to its AP curve. If, as the firm hires more labor up to 3.5 workers a day, its average product rises, its average variable cost falls. If average product is at a maximum, average variable cost is at a minimum. If as the firm hires more labor its average product diminishes, its average variable cost rises.

An increase in wage rates or some other component of *variable* cost shifts the variable cost curves (TVC and AVC) and the marginal cost curve (MC) upward but leaves the fixed cost curves (AFC and TFC) unchanged. So, for example, if the interest expense paid by a trucking company increases, the fixed cost of transportation services increases, but if the wage rate paid to truck drivers increases, the variable cost and marginal cost of transportation services increase.